Maximum Mark: 10 **Quiz-3** **Time: 55 minutes**

1. (points) [1+2+2]
   (a) Write the ground state electronic configuration for atoms with atomic number 10 and 19 respectively.
   (b) What will be the L, S and J values of the ground state in the above case (part (a)). Briefly mention the reason for the L, S, J values and write the final state term in spectroscopic notation (with multiplicity).
   (c) Calculate the degeneracy of the C_6 atom (ground state) and list all the quantum numbers describing the possible states explicitly (may use Table to list these parameters).

2. (points) [3] For a 2-electron atom, write the proper spatial wave functions (general state) and using this, derive the final expression (*best possible reduced form with ALL STEPS shown explicitly*) for the first order energy correction associated with the exchange term.

3. (points) [2] For a 2-electron system, construct a simple trial wave function for the 2^1S state corresponding to the configuration 1s2s (*may use generic description e.g.* u_{nl}, *etc.* rather than explicit form with a normalisation factor).

Useful Results

- $\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} e^{(-r/a_0)}$;
  $\psi_{200} = \frac{1}{2\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(1 - \frac{r}{2a_0}\right) e^{(-r/2a_0)}$;